
UNIT 5 DATA PREPARATION FOR ANALYSIS

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5.0 INTRODUCTION

Having established a strong foundation in the fundamentals of data science, types of data, and data acquisition tools and techniques through Block-1 (Units 1 to 4), learners are now prepared to move into the next crucial phase of the data science pipeline: preparing data for analysis. Block-1 equipped students with the knowledge to identify, classify, and collect data from diverse sources using appropriate tools and methodologies. However, real-world data is often messy, incomplete, inconsistent, or redundant—posing significant challenges before any meaningful analysis can be conducted.

Unit 5, “Data Preprocessing,” builds directly upon this foundation by introducing the essential techniques required to transform raw data into a structured, clean, and analysis-ready format. It emphasizes the importance of data cleaning, integration, transformation, and reduction—key processes that ensure data quality and consistency. These tasks are not just technical necessities but critical enablers of trustworthy analysis and reliable decision-making.

The unit also introduces the concept of data curation and its significance in maintaining the integrity and usability of datasets over time. By learning how

to curate and prepare data, students develop practical skills that bridge the gap between data collection (explored in Block-1) and deeper statistical or machine learning tasks that follow in later units.

This progression from data understanding and acquisition to data preprocessing reflects the real-world workflow of a data scientist or analyst. It reinforces the idea that high-quality input data is the backbone of accurate and insightful analytics, making Unit 5 a natural and essential continuation of the learning journey that began in Block-1.

5.1 OBJECTIVES

After finishing this unit, you will be able to:

- Describe the meaning of "data quality".
- Explain basic techniques for data preprocessing.
- Use the technique of data selection and extraction.
- Define data curation and data integration.
- Describe the knowledge discovery.

5.2 NEED FOR DATA PREPARATION

In the present time, data is one of the key resources for a business. Any raw facts are known as data and are processed to create information; information is integrated to create knowledge. Since knowledge is power, it has evolved into a modern currency, which is valued and traded between parties. Everyone wants to discuss the knowledge and benefits they can gain from it. Data is the most significant resource available to marketers, agencies, publishers, media firms, and others. But only high-quality data is useful. A good quality data set is suitable for decision-making. How do you gauge the data quality? The usefulness of the data for the intended purpose and its completeness, accuracy, timeliness, consistency, validity, and uniqueness are used to determine the data's quality.

In simpler terms, data quality refers to how accurate and helpful the data are for the task at hand. Further, data quality also relates to the actions that apply the necessary quality management procedures and methodologies to make sure the data is useful and actionable for the data consumers. A wide range of elements, including accuracy, completeness, consistency, timeliness, uniqueness, and validity, influence data quality. Figure 1 shows the basic factors of data quality.



Figure 5.1: Factors of Data Quality

These factors are explained below:

- **Accuracy** - The data must be true and reflect events that take place in the real world. Accuracy measures determine how closely the figures agree with the verified right information sources.
- **Completeness** - The degree to which the data is complete determines how well it can provide the necessary value.
- **Consistency** - Data consistency is the homogeneity of the data across applications, networks. Data consistency is most desirable when the data is gathered from several sources. For example, identical datasets should not conflict if they are stored in different locations.
- **Timeliness** - Data that is timely is readily available whenever it is needed. The timeliness factor also entails keeping the data accurate to make sure it is always available, accessible, and updated in real time.
- **Uniqueness** - Uniqueness is defined as the lack of duplicate or redundant data across all datasets. The collection should contain zero duplicate records.
- **Validity** - Data must be obtained in compliance with the firm's defined business policies and guidelines. The data should adhere to the appropriate and standard formats. In addition, all the dataset values should be within the specified ranges.

Consider yourself a manager at a company, say XYZ Pvt Ltd, who has been asked to research the sales statistics for a specific organization, say ABC. You immediately get to work on this project by carefully going through the ABC company's database and data warehouse for the parameters or dimensions (such as the products, prices, and units sold etc.), which may be used in your study. However, your enthusiasm suffers a major problem when you see that several of the attributes for different tuples do not have any recorded values. You want to incorporate the information in your study on whether each item purchased was marked in the related purchase records, but you find that this data has not been recorded. According to users of this database system, the data recorded for some transactions contained mistakes; therefore, such transactions were causing anomalies in the data. Thus, this data was not registered.

The three characteristics of data quality—accuracy, completeness, and consistency—are highlighted in the paragraph above. Large databases and data warehouses used in the real world frequently contain inaccurate, incomplete, and inconsistent data. What may be the causes of such erroneous data in the databases? There may be problems with the data collection tools, which may result in mistakes during the data-entry processes. Some of these problems can be due to the following reasons:

- Personal biases: There can be instances when users do not want to submit their personal information. They may purposefully enter inaccurate data values for required fields in such cases. For example, if a user is asked to enter his/her date of birth, a person may select the default date as his/her birthdate (please note in many applications, the default date may be "January 1"). This inaccurate data is referred to as *disguised missing data*.
- Data transfer errors: Data transfer errors may be generated during data communication due to certain technical limitations. For example, the use of synchronized data transit and consumption may be constrained by technological limitations, such as a short buffer capacity.
- The unreliable data may result from differences in naming conventions, data codes, or input field types (e.g., the format of date fields such as dd/mm/yy or mm/dd/yyyy) etc.
- Finally, cleansing of data may be needed to remove duplicate tuples.

Incomplete data can be caused by a variety of circumstances. Certain properties, such as customer information for sales transaction data, might not always be accessible. It is likely that some data might be omitted since it was not thought to be important at the time of input. A misinterpretation or malfunctioning technology could prevent the recording of essential data. For example, the data, which did not match the previously stored data, may be eliminated. Furthermore, it is likely that the data's past alterations or histories are not documented. For tuples with missing values for some properties, it could be required to infer missing data.

5.3 DATA PREPROCESSING

Preprocessing is the process of taking raw data and turning it into information that may be used for data analysis. Data cleaning, data integration, data reduction and data transformation are the main phases of data preprocessing (see Figure 2). In addition, data discretization is another component of data preprocessing. You may refer to the further readings for more details on data discretization.

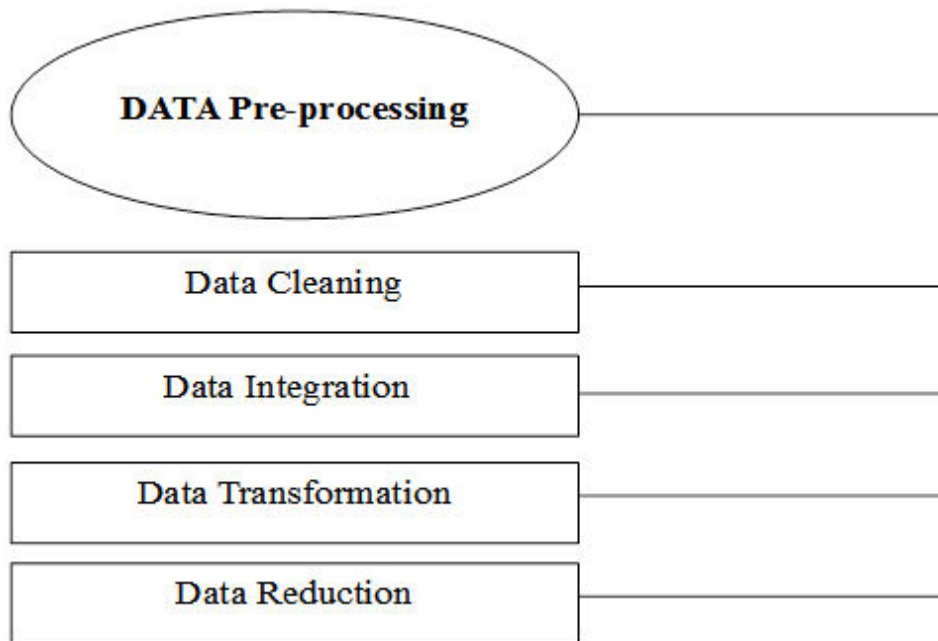


Figure 5.2: Data preprocessing

5.3.1 Data Cleaning

Data cleaning is an essential step in data preprocessing. It is also referred to as scrubbing. It is crucial for the construction of a good analysis model. Data cleaning is a required but frequently overlooked aspect of data preprocessing. Real-world data typically exhibit incompleteness, noise, and inconsistency. In addition to addressing discrepancies, this task entails filling in missing numbers, smoothing out noisy data, and eliminating outliers. Generally, a good data cleaning process helps reduce errors in data modeling and enhances data quality. Although it might be a time-consuming and laborious operation, it is necessary to fix data inaccuracies and delete bad entries. In the subsequent paragraphs, we discuss some basic data cleansing operations.

a. Missing Values

Consider you need to study *Customer and Sales* data for ABC Company. You examined the data and pointed out that numerous tuples lack recorded values for several characteristics or attributes (for example, customer income). The following techniques can be used to add values that are lacking.

- i) **Ignore the tuple:** Consider that a data science task involves classification based on the attribute(s) whose value is/are missing. In such a case, ignoring the tuples with missing classifying attribute(s) values would be desirable. This method is particularly detrimental when each attribute has a significantly different percentage of missing values. In such a case, we disregard the remaining characteristics or attributes in the tuple.
- ii) **Manually enter the omitted value:** In general, this strategy is time-consuming and might not be practical for huge data sets with a substantial number of missing values.

- iii) **Fill up the blank with a global constant:** A single constant, such as "Unknown" or " $-\infty$ ", should be used to replace all missing attribute values. If missing data are replaced with "Unknown", the analysis algorithm can mistakenly think that all "Unknown" values collectively comprise a valid data category. So, despite being simple, this strategy is not perfect.
- iv) **To fill in the missing value, use a measure of the attribute's central tendency (such as the mean or median):** The median should be used for skewed data distributions, whereas the mean can be used for normal (symmetric) data distribution. Assume, for instance, that the ABC company's customer income data distribution is symmetric and that the mean income is INR 50,000/-. Use this value to fill in the income value that is missing.
- v) **For all samples that belong to the same class as the specified tuple, use the mean or median:** For instance, if we were to categorize customers based on their credit risk, the mean income value of customers who belonged to the same credit risk category as the given tuple, might be used to fill in the missing value. If the data distribution is skewed for the relevant class, it is best to utilize the median value.
- vi) **Fill in the blank with the value that is most likely to be there:** This result can be reached using regression, inference-based techniques using a Bayesian formalization, or decision tree induction. For example, using the other characteristics of your customer's data, you may create a decision tree to forecast the missing income of a specific customer.

b. **Noisy Data**

Noise is the variance or random error in a measured variable. Noise can be an outlier, which can be recognized by employing tools for data visualization and basic statistical description techniques such as scatter plots and boxplots. How can the data be "smoothed" out to reduce noise if the given data has numeric property (for example, the price of the items)? The following are some of the data-smoothing strategies.

- i) **Binning:** Binning techniques smooth sorted data values by looking at their "neighborhood" or nearby values. The values, which have been sorted, are divided into various "buckets" or bins. Binning techniques carry out local smoothing since they look at the data values' surroundings. Smoothing by bin means that each value in a bin is changed to the mean value of that bin. As an illustration, suppose a bin contains three numbers 5, 10 and 12. The average of these three numbers in the bin is 9. Consequently, the value nine replaces each of the bin's original values.

Similarly, smoothing by bin medians, which substitutes the bin median for each bin value, can be used. In addition, you can use the maximum or the minimum values in a bin, also referred to as bin boundaries of a specific bin, in place of bin mean or median. This type of smoothing is called smoothing by bin boundaries. This

method uses the nearest boundary value to replace each bin value. In general, the smoothing effect increases with increasing breadth of bin boundaries. The bins can also be created using identical intervals, leading to constant ranges of values.

- ii) **Regression:** Regression is a method of mapping the data values to a function. This regression function may then be used to smooth out the data. Finding the "best" line to fit two traits (or variables) is the goal of linear regression, which enables one attribute to predict the other. As an extension of linear regression, multiple linear regression involves more than two features and fits the data to a multidimensional surface.
- iii) **Outlier analysis:** Clustering groups comparable values into "clusters". These clusters can then be used to identify outliers. It makes sense to classify values that are outliers as being outside the set of clusters.
- iv) **Data discretization:** Data discretization is a data transformation and reduction technique. It is an extensively used data smoothing technique. Data discretization decreases the number of distinct values for each property using some technique; for instance, you can discretize.
- v) **Data using the binning approach:** Concept hierarchies are a data discretization technique that can also be applied to smooth out the data. The quantity of data values that the analysis process must process is decreased by a concept hierarchy. For example, the price variable, which represents the price value of commodities, may be discretized into "lowly priced", "moderately priced", and "expensive" categories.

Steps of Data Cleaning

Data cleaning is a critical step in the data preparation process that helps ensure data quality and accuracy. While specific data cleaning steps may vary depending on the nature of the data and the organization's needs, here are some basic steps to create a framework for data cleaning.

1. Remove duplicate or irrelevant observations: Duplicate data may be produced when data sets from various sources are combined, scraped, or data is obtained from clients or other departments.
2. Fix structural errors: When measuring or transferring data, you may come across structural mistakes such as unusual naming practices, typographical errors, or wrong capitalization. Such inconsistencies may lead to mislabeled categories or classes. For instance, "N/A" and "Not Applicable", which might be present on any given document, may create two different classifications. Instead, they should be considered under the same heading or should be considered as missing values.
3. Managing unwanted outliers: Outliers can cause the formation of inaccurate models; for example, outliers can result in the construction of an incorrect regression function. However, decision tree models are more

robust to outliers than linear regression. We should not eliminate outliers unless there is a compelling reason to do so. Sometimes removing them can improve performance, but not always. Therefore, the outlier must be eliminated for a good cause, such as suspicious measurements that are unlikely to be present in real-life data.

4. **Handling missing data:** Missing data is a deceptively difficult issue in machine learning. We cannot just ignore or remove the missing observations. They must be carefully treated since they can indicate a serious problem. Data gaps resemble puzzle pieces that are missing. Dropping it is equivalent to denying that the puzzle slot is there. It is like trying to put a piece from another puzzle into this one. Furthermore, you need to be aware of how you report missing data. Instead of just filling it in with the mean, you can effectively let the computer choose the appropriate constant to account for missingness by using this flagging and filling method.
5. **Validate and QA:** You should be able to respond to these inquiries as part of fundamental validation following the data cleansing process, for example:
 - Does the data make sense?
 - Does the data abide by the regulations that apply to its particular field?
 - Does it support or refute your hypothesis? Does it offer any new information?
 - Can you see patterns in the data that will support your analysis?
 - Is there a problem with the data quality?

Methods of Data Cleaning

The following are some of the methods of data cleaning. Some of the methods have already been discussed as steps of data cleaning.

1. **Ignore the tuples:** This approach is not particularly practical because it can only be used when a tuple has multiple characteristics and missing values.
2. **Fill in the missing value:** This strategy is neither practical nor very effective. Additionally, the process could take a long time. The missing value must be entered in the approach. The most common method for doing this is manually putting the value for the missing value. Other options include using the attribute mean or the value with the highest probability for the missing value.
3. **Binning method:** The Binning method, as explained earlier, creates bins of nearby values. These values, then, are used to smooth the data.
4. **Regression:** The regression function can predict and smooth the data for machine learning. You may use bivariate or multivariate linear regression for this purpose.

5. **Clustering:** The primary objective of the clustering approach is to group the comparable values in a cluster, thus, creating data clusters. Clustering also aids in ascertaining possible outliers.

5.3.2 Data Integration

Data from many sources, such as files, data cubes, databases (both relational and non-relational), etc., may be combined before a machine learning algorithm can use it as training or test data. The data from the sources may have the following characteristics:

- The data sources may be homogeneous or heterogeneous.
- The data sources may contain structured, unstructured, or semi-structured data.

Redundancies and inconsistencies can be reduced and avoided with careful integration. The following are some of the issues of data integration.

- a. **Entity identification problem:** In many projects, data from several sources are integrated into a consistent data set. For example, a data warehouse gathers data from several sources into coherent data storage. These sources include various databases, data cubes, and flat files.

During data integration, there are several things to consider. Integration of schemas and object matching might be challenging. How is it possible to match comparable real-world things across different data sources? The entity identification problem describes this. How can a computer or data analyst be sure that a client's ID in one database and their customer number in another database refer to the same attribute? Examples of metadata for each attribute include the name of the attribute, its definition, data type, permissible range of values and null rules for handling empty, zero, or null values. Such metadata can be used to avoid errors during the integration of the schema. The data may also be transformed with the aid of metadata. For example, in two different instances of data of an organization, the code for pay data might be "H" for high income and "S" for small income in one instance of the database; the same pay code in another instance of a database may be 1 and 2.

When comparing attributes from one database to another during integration, the data structure must be considered properly. This is done to make sure that any referential constraints and functional dependencies on attributes present in the source system are also present in the target system. For instance, one system might apply a discount to the entire order but only to certain items by another. If this is not found before integration, things in the target system can be incorrect.

- b. **Redundancy and Correlation Analysis:** Another crucial problem in data integration is redundancy. If an attribute (like annual income, for example) can be "derived" from another attribute or group of data, it may be redundant. Inconsistent attributes or dimension names can also bring redundancies in the final data set.

Correlation analysis can identify some redundancies. Based on the available data, such analysis can quantify the strength of relationships between two attributes. We employ the chi-square test (χ^2) for finding relationships between nominal data. Numeric attributes can be analyzed using the correlation coefficient and covariance, which look at how one attribute's values differ from those of another.

- c. **Tuple duplication:** Duplication should be identified at the tuple level, e.g., when, for a particular unique data entry case, there are two or more identical tuples. Additional data redundant sources include – deformalized tables, which are frequently used to increase performance by reducing joins; faulty data entry or updating some (not all) redundant data occurrences, etc. Inconsistencies frequently appear between different duplicates. For example, there may be inconsistency as the same purchaser's name may appear with multiple addresses within the purchase order database. This might happen if a database for purchase orders has attributes for the buyer's name and address rather than a foreign key to this data.
- d. **Data value conflict detection and resolution:** Data value conflicts must be found and resolved as part of data integration. As an illustration, attribute values from many sources may vary for the same real-world thing. Variations in representation, scale, or encoding may be the cause of this. In one system, a weight attribute might be maintained in British imperial units, while in another, metric units. For a hotel chain, the cost of rooms in several cities could include various currencies, services (such as a complimentary breakfast) and taxes. Similarly, every university may have its own curriculum and grading system. When sharing information among them, one university might use the quarter system, provide three database systems courses, and grade students from A+ to F, while another would use the semester system, provide two database systems courses, and grade students from 1 to 10. Information interchange between two such universities is challenging to establish accurate course-to-grade transformation procedures. An attribute in one system might be recorded at a lower abstraction level than the "identical" attribute in another since the abstraction level of attributes might also differ. As an illustration, an attribute with the same name in one database may relate to the total sales of one branch of a company, however, the same attribute name in another database can refer to the company's overall regional shop sales.

5.3.3 Data Reduction

In this phase, data is trimmed. The number of records, attributes, or dimensions can be reduced. When reducing data, one should keep in mind that the outcomes from the reduced data should be identical to those from the original data. Consider that you have chosen some data for analysis from *ABC Company's* data warehouse. The data set will probably be enormous! Large-scale complex data analysis and mining can be time-consuming, rendering such a study impractical or unfeasible. Techniques for data reduction can be applied to create a condensed version of the data set that is

considerably smaller while meticulously retaining the integrity of the original data. In other words, mining the smaller data set should yield more useful results while effectively yielding the same analytical outcomes. This section begins with an overview of data reduction tactics and then delves deeper into specific procedures. Data compression, dimensionality reduction, and numerosity reduction are data reduction methods.

- a. **Dimensionality reduction** refers to the process of lowering the number of random variables or qualities. Principal components analysis and wavelet transformations are techniques used to reduce data dimensions by transforming or rescaling the original data. By identifying and eliminating duplicated, weakly relevant, or irrelevant features or dimensions, attribute subset selection is a technique for dimensionality reduction.
- b. **Numerosity reduction** strategies substitute different, more compact forms of data representation for the original data volume. Both parametric and non-parametric approaches are available. In parametric techniques, a model is employed to estimate the data, which frequently necessitates the maintenance of only the data parameters rather than the actual data. (Outliers may also be stored.) Examples include log-linear models and regression. Non-parametric methods include the use of histograms, clustering, sampling, and data cube aggregation to store condensed versions of the data.
- c. **Transformations** are used in **data compression** to create a condensed or "compressed" version of the original data. Data compression can be - lossless data compression, in which the original data can be recovered from the compressed data without losing any information, or lossy data compression, in which only a fraction of the original data can be recovered from the compressed data on decompression. There are a number of lossless string compression algorithms; however, they typically permit only a small amount of data manipulation. Techniques for reducing numerosity and dimensions can also be categorized as data compression methods.

There are other additional data reduction techniques. You may please note that the data reduction techniques also require computational time. The time saved by analysis on a smaller data set should not be "erased" or outweighed by the computational effort required for data reduction.

Data Discretization: - It is regarded as a component of data reduction. The data discretization alters numerical data by converting values to intervals or concept labels. These techniques enable data analysis at various levels of granularity by automatically generating concept hierarchies for the data. Binding, histogram analysis, decision tree analysis, cluster analysis, and correlation analysis are examples of discretization techniques. Concept hierarchies for nominal data may be produced based on the definitions of the schema and the distinct attribute values for every attribute.

5.3.4 Data Transformation

This procedure is used to change the data into formats that are suited for the analytical process. Data transformation involves transforming or consolidating the data into analysis-ready formats. The following are some data transformation strategies:

- a. **Smoothing:** Smoothing is a process which attempts to reduce data noise. You can use methods like binning, regression, and grouping for data smoothening.
- b. **Attribute construction (or feature construction):** Attribute construction is the process of constructing additional attributes using the set of data attributes. The primary objective of attribute construction is to aid the analysis process,
- c. **Aggregation:** Aggregation is the process of summarizing the data of an attribute based on some criteria; for instance, the daily sales data may be combined to produce monthly or yearly sales. This process is often used to build a data cube for data analysis at different levels of abstraction.
- d. **Normalizations:** Normalization resizes attribute data to fit a narrower range: -1.0 to 1.0 ; or 0.0 to 1.0 .
- e. **Discretization:** Discretization is the process where interval labels replace the raw values of a numeric attribute (e.g., age) (e.g., 0–10, 11–20, etc.) or conceptual labels (e.g., *youth*, *adult*, *senior*). A concept hierarchy for the number attribute can then be created by recursively organizing the labels into higher-level concepts. To meet the demands of different users, more than one concept hierarchy might be built for the same attribute.
- f. **Concept hierarchy creation using nominal data:** These concept hierarchies allow for extrapolating higher-level concepts, for example - a street to concepts like a city or a country. At the schema definition level, numerous hierarchies for nominal qualities can be automatically created and are implicit in the database structure.

Check Your Progress 1:

1. What is meant by data preprocessing?
2. Why is preprocessing important?
3. What are the 5 characteristics of data processing?
4. What are the major steps of data preprocessing?
5. What is data cleaning?
6. What is the importance of data cleaning?
7. What are the main steps of Data Cleaning?

5.4 SELECTION AND DATA EXTRACTION

The process of choosing the best data source, data type, and collection tools is known as data selection. Prior to starting the actual data collection procedure, data selection is conducted. This concept makes a distinction between selective data reporting (excluding data that is not supportive of a study premise) and active/interactive data selection (using obtained data for monitoring activities/events or conducting secondary data analysis). Data integrity may be impacted by how acceptable data are selected for a research project.

The main goal of data selection is to choose the proper data type, source, and tool that enables researchers to solve research issues effectively. This decision typically depends on the research area, research questions, prior research, and the availability of the data sources.

When cost and convenience considerations make you decide which "appropriate" data to collect, then you may face data integrity issues. Cost and convenience are unquestionably important variables to consider while deciding. However, researchers should consider how much these factors can skew the results of their study.

Data Selection Issues

When choosing data, researchers should be conscious of a few things, including:

- Researchers should select the right type and appropriate data sources to respond appropriately to the stated research questions viz, appropriate methods for obtaining representative samples and appropriate tools for gathering data. It is difficult to separate the choice of data type and source from the tools used to get the data. The type/source of data and the procedures used to collect the data should be compatible.

Types and Sources of data: Different data sources and data types can be expressed in a variety of ways. The following are the two main categories of data types:

- Quantitative data are expressed as numerical measurements at the interval and ratio levels.
- Qualitative data can take the form of text, images, music, and video.

Although preferences within scientific disciplines differ as to which type of data is preferred, some researchers employ information from both quantitative and qualitative sources to comprehend a certain event more thoroughly.

Researchers get data from people that may be qualitative (such as by studying child-rearing techniques) or quantitative (biochemical recording markers and anthropometric measurements). Field notes, journals, laboratory notes, specimens, and first-hand observations of people, animals, and plants can all be used as data sources.

Choosing the right data is discipline-specific and primarily influenced by the purpose of investigation, the body of prior research, and the availability of data sources. The following list of questions will help you choose the right data type and sources:

- What is the research question?
- What is the investigation's field of study? (This establishes the guidelines for any investigation. Selected data should not go beyond what is necessary for the analysis.)
- According to the literature (prior research), what kind of information should be gathered?
- Which form of data—qualitative, quantitative, or a combination of both—should be considered?

Data extraction is the process of gathering or obtaining many forms of data from numerous sources, many of which may be erratically organised or wholly unstructured. Consolidating, processing and refining data enables information to be kept in a centralized area so that it can be altered. These data storage units could be local, online, or a combination of the two.

Data Extraction and ETL

To appreciate the importance of data extraction, let us briefly define the ETL process.

1. **Extraction:** One or more sources or systems are used to collect the data. During the extraction phase, relevant data is located, identified, and then prepared for processing or transformation. One can finally analyse them for business knowledge by combining various data types through extraction.
2. **Transformation:** Once the data has been effectively extracted, it can be refined further. Data is cleansed, sorted, and structured as part of the transformation process. Duplicate entries will be removed, missing values will be filled in or removed, and audits will be performed to offer reliable, consistent, and usable data.
3. **Loading:** The high-quality data, which is generated by the transformation process, is subsequently sent to a single, centralized target location for storage and analysis.

Data extraction tools

The tools listed below can be used for tasks other than simple data extraction. The features of these software are given below:

1. **Scrape storm:** Scrape storm is software that uses AI to scrape the web or gather data. It is compatible with Windows, Mac, or Linux operating systems and has a simple and straightforward visual operation. This program automatically detects objects like emails, numbers, lists, forms, links, photos, and prices. When transferring extracted data to Excel, MongoDB, CSV, HTML, TXT, MySQL, SQL Server, PostgreSQL, Google Sheets, or WordPress, it can make use of a variety of export strategies.

2. **Altair Monarch:** Monarch is a desktop-based self-service and does not require any programming. It can link to various data sources, including big data, cloud-based data, and both organised and unstructured data. It connects to, cleans, and processes data with high speed and no errors. It employs more than 80 built-in data preparation functions. When you use this tool, you waste less time making data legible; therefore, you can spend more time creating higher-level knowledge from the data.
3. **Klippa:** Klippa can process contracts, invoices, receipts, and passports over the cloud. For most documents, the conversion time taken by the software may be between one and five seconds. The data classification and manipulation may be done online, round-the-clock. It supports a variety of file types, including PDF, JPG, and PNG. It can also convert between JSON, PDF/A, XLSX, CSV, and XML. Additionally, the software handles file sharing, custom branding, payment processing, cost management, and invoicing management.
4. **NodeXL:** Basic version of the software is a free, open-source add-on extension for Microsoft Excel 2007, 2010, 2013, and 2016. Since the software is an add-on, data integration is not performed; instead, it focuses on social network analytics. Advanced network analytics, text and sentiment analysis, and robust report generating are extra capabilities included with NodeXL Pro.

Check Your Progress 2:

1. What is the data selection process?

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2. What is Data Extraction and define the term ETL?

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3. What are the challenges of data extraction?

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5.5 DATA CURATION

In the previous sections, we have discussed data preprocessing. The objective of data preprocessing is to obtain high-quality data for analysis. In this section, we discuss data curation. Data curation is integrating and maintaining high-quality data for a specific purpose in an organization using multiple data sources. Data curation aims to create high-quality data sets that can be accessed and used by data users, who may include company

employees or any other person looking for such information. Data curation involves collecting the data from its sources; integrating and arranging the collected data; indexing the information so generated; and categorizing the information to support business decisions, academic needs, scientific research, etc. Data curation is a step in the more extensive data management process that helps prepare data sets for usage in business intelligence (BI) and analytics applications.

5.5.1 Steps of Data Curation

There are numerous tasks involved in curating data sets, which can be divided into the following main steps.

- The data that will be required for the proposed analytics applications should be determined.
- Map the data sets and note the metadata that goes with them.
- Collect the data sets.
- The data should be ingested into a system, such as, a data lake, a data warehouse, etc.
- Clean the data to remove abnormalities, inconsistencies, and mistakes, including missing values, duplicate records, and spelling mistakes.
- Model, organize, and transform the data to prepare it for specific analytics applications.
- To make the data sets accessible to users, create searchable indexes.
- Maintain and manage the data in compliance with the requirements of continuous analytics and the laws governing data privacy and security.

5.5.2 Importance of Data Curation

The following are the reasons for performing data curation.

1. **Helps to organize pre-existing data for a corporation:** Businesses produce a large amount of data on a regular basis. For example, when a customer clicks on a website, adds something to their cart, or completes a transaction, an online retailer might record that information. Data curators assist businesses in better understanding vast amounts of information by integrating and aggregating the data into data sets.
2. **Connects professionals in different departments:** When a company engages in data curation, it often brings together people from several departments who might not typically collaborate. Data curators might collaborate with stakeholders, system designers, data scientists, and data analysts to collect and transfer information.
3. **High-quality data:** It typically uses organizational techniques that make it simple to grasp and have fewer errors. Curators may make sure that a company's research and information continue to be of the highest caliber because the data curation process entails cleansing the data. Removing unnecessary information may facilitate better data set structure.
4. **Makes data easy to understand:** Data curators make sure there are no

errors and utilize proper formatting. This makes it simpler for specialists, who are not knowledgeable about a research issue, to comprehend a data set.

5. **Allows for higher cost and time efficiency:** A business may spend more time and money organizing and distributing data if it does not regularly employ data curation. Because prior data is already organized and distributed, businesses that routinely do data curation may be able to save time, effort, and money. Businesses can reduce the time to obtain and process data by using data curators.

Check Your Progress 3:

1. What is the Importance of Data Curation?

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2. Explain Data Curation.

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3. What are the goals of data curation?

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4. What are the benefits of data curation?

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5.6 DATA INTEGRATION

Data Integration creates coherent data storage by combining data from several sources. Smooth data integration is facilitated by the resolution of semantic heterogeneity, metadata, correlation analysis, tuple duplicate identification, and data conflict detection. It is a tactic that combines data from several sources so that data consumers may access it in a single, consistent view that displays the required data. Systems can communicate using flat files, data cubes, or numerous databases. Data integration is crucial because it maintains data accuracy while providing a consistent view of dispersed data. It helps the analysis tools extract valuable information, which in turn helps the executive and management make tactical choices that will benefit the company.

5.6.1 Data Integration Techniques

Manual Integration-When integrating data, this technique avoids employing automation. The data analyst gathers, purifies, and integrates the data to create actionable data. A small business with a modest amount of data can use this approach. Nevertheless, manual data integration, which is extensive, complex, and ongoing, will take a lot of time.

Middleware Integration-Data from many sources are combined, normalized, and stored in the final data set using middleware software. This method is employed when an organization integrates data from historical systems into modern systems. Software called middleware serves as a translator between outdated and contemporary systems. You may be required to connect the interfaces of the two systems. Such integration is designed for specific systems only.

Application-based Integration- It uses software applications to extract, transform, and load data from various sources. Although this strategy saves time and effort, it is a little more difficult because creating such an application requires technical knowledge.

Uniform Access Integration- This method integrates information from a wider range of sources. In this instance, however, the data is left in its initial place and is not moved. To put it simply, this technique produces a unified view of the combined data. The integrated data does not need to be saved separately because the end user only sees the integrated view.

5.6.2 Data Integration Approaches

There are two basic data integration approaches. These are –

Tight Coupling- It combines data from many sources into a single physical location using ETL (Extraction, Transformation, and Loading) tools.

Loose Coupling- The source databases are the most efficient place to store facts with loose coupling. This method offers an interface that receives a user query, converts it into a format that the source databases can understand, and immediately transmits the question to the source databases to get the answer.

Check Your Progress 4:

1. What is meant by data integration?

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2. What is an example of data integration?

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3. What is the purpose of data integration?

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5.7 KNOWLEDGE DISCOVERY

Knowledge discovery in databases is the process of obtaining pertinent knowledge from a body of data (KDD). This well-known knowledge discovery method includes several processes, such as data preparation and selection, data cleansing, incorporating prior knowledge about the data sets, and interpreting precise answers from the observed results.

Marketing, fraud detection, telecommunications, and manufacturing are some of the key KDD application areas. In the last ten years, the KDD process has reached its pinnacle. Inductive learning, Bayesian statistics, semantic query optimisation, knowledge acquisition for expert systems, and information theory are just a few of the numerous discovery-related methodologies. The ultimate objective of KDD is to extract high-level knowledge from low-level data. Due to the accessibility and quantity of data available today, knowledge discovery is a challenge of astounding importance and necessity. Given how swiftly the topic has expanded, it is not surprising that professionals and experts today have access to a variety of treatments of data.

Steps of Knowledge Discovery

1. **Developing an understanding of the application domain:** Knowledge discovery starts with this preliminary step. It establishes the framework for selecting the best course of action for a variety of options, such as transformation, algorithms, etc. The individuals in charge of a KDD project need to be aware of the end users' goals as well as the environment in which the knowledge discovery process will take place.
2. **Selecting and producing the data set that will be used for discovery:** Once the objectives have been specified, the data that will be used for the knowledge discovery process should be identified. Some of the basic functions for this step are - determining what data is accessible, identifying what information is expected to be obtained from the data, and combining all the data for knowledge discovery into one data set, etc. Knowledge discovery is important since it extracts knowledge and insight from the given data. This provides the framework for building the models.
3. **Preprocessing and cleansing:** This step helps in increasing the data reliability. For instance, the outcome of the Data Mining supervised approach may be considered not dependable if it is established that a specific attribute is unreliable or has a sizable amount of missing data. Preprocessing comprises data cleaning, which includes handling the missing quantities and removing noise or outliers. Such data cleansing might make use of sophisticated statistical methods or analysis algorithms. After developing a prediction model for these features, missing data can be forecasted.

4. **Data Transformation:** This phase entails creating and preparing the necessary data for knowledge discovery. Here, techniques of attribute transformation (such as discretization of numerical attributes, functional transformation etc.) and dimensionality reduction (such as feature selection, feature extraction, record sampling etc.) are employed. This step, in general, is different for every project. However, this step is essential for the success of the KDD project, as a proper transformation results in proper analysis and proper conclusions.
5. **Prediction and description:** Once data transformation is complete, you can decide to analyze data using classification, regression, clustering, or any other method based on KDD objectives. A forecast and a description are the two main objectives of knowledge discovery. The visualization aspects are included in descriptive knowledge discovery. Most knowledge discovery techniques use inductive learning, which generalizes enough prepared models to produce a model either explicitly or implicitly. The fundamental premise of the inductive technique is that the trained model holds for the examples that follow.
6. **Choosing a Knowledge Discovery Algorithm:** After determining the approach, the next step involves selecting appropriate strategies. At this stage, a specific technique needs to be applied to uncover patterns. When comparing precision and understandability, neural networks enhance the former, while decision trees improve the latter. There are multiple ways in which each meta-learning system could achieve success. The objective of meta-learning is to elucidate why a data analysis algorithm succeeds or fails in solving a specific problem. Consequently, this methodology aims to comprehend the conditions under which a data analysis algorithm is most efficient. Each algorithm encompasses parameters and learning techniques, such as tenfold cross-validation or alternative divisions for training and testing.
7. **Implementing the Data Analysis Algorithm:** Ultimately, the data analysis algorithm is deployed. The approach might require multiple iterations before yielding a desirable outcome. For example, by varying the algorithms, you can modify variables such as the minimum number of instances in a single decision tree leaf.
8. **Evaluation:** In this phase, the patterns, principles, and reliability of the outcomes from the knowledge discovery process are evaluated and interpreted in the context of the objective established in the previous step. At this point, we examine the preprocessing steps and their influence on the results. For instance, you might introduce an additional feature in step 4 and then proceed. The key focal points in this stage are the comprehensibility and effectiveness of the generated model. Additionally, any identified knowledge is documented during this stage for future reference.

Check Your Progress 5:

1. What is Knowledge Discovery?

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2. What are the Steps involved in Knowledge Discovery?

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3. What are knowledge discovery tools?

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4. Explain the process of KDD.

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5.8 SUMMARY

Despite the development of several methods for preparing data, the intricacy of the issue and the vast amount of inconsistent or unclean data mean that this field of study is still very active. This unit gives a general overview of data preprocessing and describes how to turn raw data into usable information. The preprocessing of the raw data included data integration, data reduction, transformation, and discretization. In this unit, we have discussed five different data-cleaning techniques that can make data more reliable and produce high-quality results. Building, organizing, and maintaining data sets is known as data curation. A data curator usually determines the necessary data sets and makes sure they are gathered, cleaned up, and changed as necessary. The curator is also in charge of providing users with access to the data sets and information related to them, such as their metadata and lineage documentation. The data curator's primary goal is to ensure users have access to the appropriate data for analysis and decision-making. Data integration is the procedure of fusing information from diverse sources into a single, coherent data store. The unit also introduced knowledge discovery techniques and procedures.

5.9 SOLUTIONS/ANSWERS

Check Your Progress 1:

1. As a part of data preparation, data preprocessing refers to any type of processing done on raw data to get it ready for a data processing technique. It has long been regarded as a crucial first stage in data mining and machine learning.
2. It raises the reliability and accuracy of the data. Preprocessing data can increase the correctness and quality of a dataset, making it more dependable by removing missing or inconsistent data values brought by human or computer mistakes. It ensures consistency in data.
3. Data quality is characterized by five characteristics: correctness, completeness, reliability, relevance, and timeliness.
4. The five major steps of data preprocessing are:
 - Data quality assessment
 - Data cleaning
 - Data transformation
 - Data reduction
5. The practice of correcting or deleting inaccurate, damaged, improperly formatted, duplicate, or incomplete data from a dataset is known as data cleaning. There are numerous ways for data to be duplicated or incorrectly categorised when merging multiple data sources.
6. Data cleansing, sometimes referred to as data cleaning or scrubbing, is the process of locating and eliminating mistakes, duplication, and irrelevant data from a raw dataset. Data cleansing, which is a step in the preparation of data, ensures that the cleaned data is used to create accurate, tenable visualizations, models, and business choices.
7. Step 1: Remove irrelevant data; Step 2: Deduplicate your data; Step 3: Fix structural errors; Step 4: Deal with missing data; Step 5: Filter out data outliers; Step 6: Validate your data.

Check Your Progress 2:

1. The process of retrieving data from the database that are pertinent to the analysis activity is known as data selection. Sometimes the data selection process comes before data transformation and consolidation.
2. Data extraction is the process of gathering or obtaining many forms of data from numerous sources, many of which may be erratically organised or wholly unstructured. The process of extracting, transforming, and loading data is called ETL. Thus, ETL integrates information from several data sources into a single consistent data store, which can be a data warehouse or data analytics system.
3. The cost and time involved in extracting data, as well as the accuracy of the data, are obstacles. The correctness of the data depends on the quality of the data source, which can be an expensive and time-consuming procedure.

Check Your Progress 3:

1. It entails gathering, organizing, indexing, and cataloging information for users within an organization, a group, or the wider public. Data curation can help with academic needs, commercial decisions, scientific research, and other endeavors.
2. The process of producing, arranging, and managing data sets so that users who are looking for information can access and use them is known as data curation. Data must be gathered, organized, indexed, and cataloged for users within an organization, group, or the broader public.
3. By gathering pertinent data into organized, searchable data assets, data curation's overarching goal is to speed up the process of extracting insights from raw data.
4. The benefits of data curation are:
 - Easily discover and use data.
 - Ensure data quality.
 - Maintain metadata linked with data.
 - Ensure compliance through data lineage and classification.

Check Your Progress 4:

1. Data integration is used to bring together data from several sources to give people a single perspective. Making data more readily available, easier to consume, and easier to use by systems and users is the foundation of data integration.
2. In the case of customer data integration, information about each customer is extracted from several business systems, such as sales, accounts, and marketing, and combined into a single picture of the client for use in customer service, reporting, and analysis.
3. Data integration combines data collected from various platforms to increase its value for your company. It enables your staff to collaborate more effectively and provide more for your clients. You cannot access the data collected in different systems without data integration.

Check Your Progress 5:

1. Knowledge discovery is the labor-intensive process of extracting implicit, unknown-before information from databases that may be beneficial.
2. Steps of Knowledge Discovery:
 - Developing an understanding of the application domain
 - Selecting and producing the data set that will be used for the discovery.
 - Preprocessing and cleansing
 - Data Transformation
 - Prediction and description
 - Deciding on a data analysis algorithm

- Utilizing the data analysis algorithm
 - Evaluation
3. The process can benefit from a variety of qualitative and quantitative methods and techniques, such as knowledge surveys, questionnaires, one-on-one and group interviews, focus groups, network analysis, and observation. It can be used to locate communities and specialists.
 4. Knowledge Discovery from Data, often known as KDD, is another commonly used phrase that is treated as a synonym for data mining. Others see data mining as just a crucial stage in the knowledge discovery process when intelligent techniques are used to extract data patterns. The steps involved in knowledge discovery from data are as follows:
 - Data cleaning (to remove noise or irrelevant data).
 - Data integration (where multiple data sources may be combined).
 - Data selection (where data relevant to the analysis task is retrieved from the database).
 - Data transformation (where data is consolidated into forms appropriate for mining by performing summary or aggregation functions, for sample).
 - Data mining (an important process where intelligent methods are applied to extract data patterns).
 - Pattern evaluation (to identify the fascinating patterns representing knowledge based on some interestingness measures).
 - Knowledge presentation (where knowledge representation and visualization techniques are used to present the mined knowledge to the user).

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